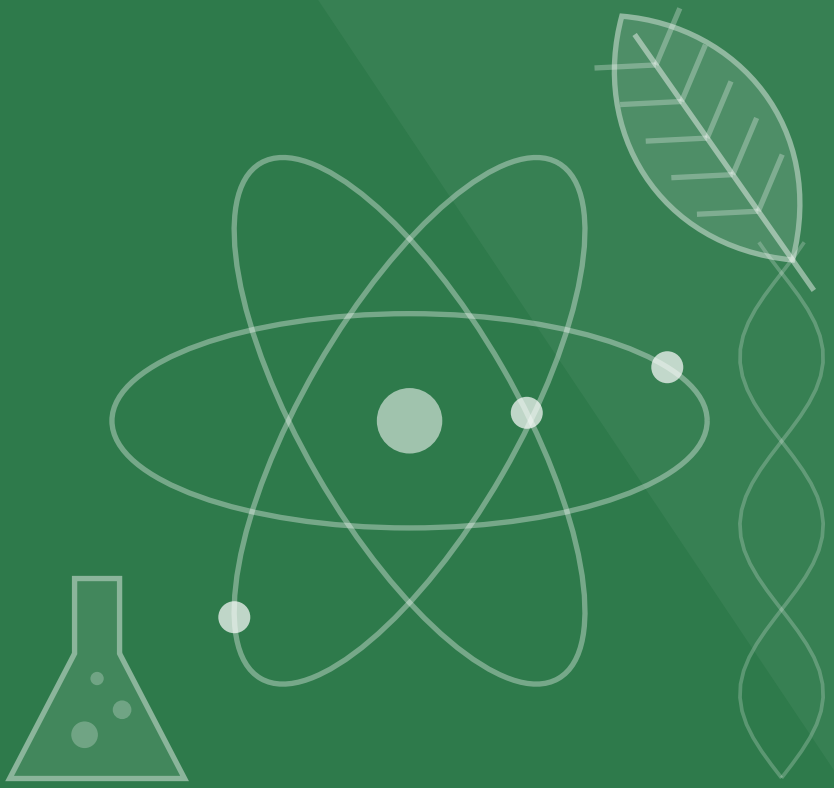
VENVLEARNING

STUDENT TEXTBOOK



# Science

GRADE 7

Lessons, examples, and practice for the whole year

Unit 1: Cycles of Matter and Energy

VENVLEARNING

# Grade 7
# Science

STUDENT TEXTBOOK

Unit 1 Cycles of Matter and Energy

A VenvLearning family homeschool curriculum

# How to use this book

Every lesson starts on a new page with the same parts, so you always know where to look. Read the Explain It boxes to see how a scientist reasons from evidence, and answer the Check Your Understanding questions. Your practice pages are in your Student Workbook, labeled with the same lesson number.

- Today's Goal (the box with the target) says what you will be able to do by the end of the lesson.
- Today's Assignment is your checklist for the day. Tick each box as you finish it.
- Each lesson page has three panels. The wide center panel is the lesson, in numbered steps: 1 Get Ready, 2 Learn, and 3 Check Your Understanding.
- The left panel holds things to remember: Key Ideas (the most important rule or fact; learn these), Watch Out warnings about common mistakes, tips, and a Remember note that sums up the lesson. Each one sits beside the part of the lesson it goes with.
- The right panel holds Words to Know (definitions), what to read in your Reader, and other extras.
- Pages that look like a page from a book, on cream paper with a brown READING tab, are things to read: stories, excerpts, and articles.
- Investigate boxes are hands-on experiments. Do them with your teacher and record what you observe.
- At the end of each week of lessons there is a short Weekly Check in your workbook. Each unit ends with the Unit Test.

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UNIT 1

# Cycles of Matter and Energy

Science practices, Earth's spheres, the water, carbon, oxygen, and nitrogen cycles, photosynthesis, respiration, food webs, and biodiversity.

Every atom in your body has been somewhere else before: in the air, the ocean, a plant, or another animal. In this unit you'll learn how scientists design experiments and explain data, then use those skills to trace how matter cycles and energy flows through Earth's systems and living things.

## In this unit you will…

- Identify variables and write CER explanations from data.
- Model the water, carbon, oxygen, and nitrogen cycles.
- Explain photosynthesis and cellular respiration as linked processes.
- Model energy flow in food webs and evaluate ways to protect biodiversity.

SCIENCE IN YOUR WORLD

Farmers, foresters, and environmental engineers use these cycles to grow food, manage forests, and protect water supplies.

UNIT VOCABULARY

| | |
|---|---|
| independent variable | The variable that is changed. |
| dependent variable | The variable that is measured. |
| photosynthesis | Light + $CO_2$ + water → glucose + oxygen. |
| cellular respiration | Glucose + oxygen → $CO_2$ + water + energy. |
| nitrogen fixation | Changing $N_2$ into usable forms. |
| biodiversity | The variety of life in an area. |

## Lessons in this unit

| Lesson | Title |
|---|---|
| 1.1 | Variables, Data, and Scientific Explanations |
| 1.2 | Earth's Spheres and the Water Cycle |
| 1.3 | The Carbon and Oxygen Cycles |
| 1.4 | The Nitrogen Cycle |
| 1.5 | Photosynthesis |
| 1.6 | Cellular Respiration |
| 1.7 | Food Webs and Energy Pyramids |
| 1.8 | Biodiversity and Ecosystem Services |

REMEMBER

LESSON 1.1
Week 1 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.1: Variables, Data, and Scientific Explanations

TODAY'S GOAL

I can identify independent, dependent, and controlled variables and write a scientific explanation using CER.

Standards: SEP 3, 4, 6

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Do the Investigate activity: Temperature and Dissolving.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–E).

## 1 Get Ready

I WONDER

Does a fizzing tablet dissolve faster in hot water or cold water? How could you find out fairly?

## 2 Learn

A controlled experiment tests how one factor affects another. The independent variable is what you change. The dependent variable is what you measure. Everything else

WORDS & MORE

WORDS TO KNOW

**independent variable**
The variable the scientist changes on purpose.

**dependent variable**
The variable that is measured; it responds to the change.

**controlled variables**
Conditions kept the same so the test is fair.

**hypothesis**
A testable prediction that explains how one variable affects another.

REMEMBER

stays the same as controlled variables. A good hypothesis is written so it can be tested: “If..., then..., because....”

| Part | Fizzing-tablet experiment |
|---|---|
| question | How does water temperature affect how long a tablet takes to dissolve? |
| hypothesis | If the water is warmer, then the tablet will dissolve faster, because particles move faster at higher temperatures. |
| independent variable | water temperature |
| dependent variable | time to dissolve (seconds) |
| controlled variables | amount of water, type and size of tablet, cup, no stirring |

WORDS & MORE

### INVESTIGATE · TEMPERATURE AND DISSOLVING

How does water temperature affect dissolving time?

You need: 3 identical effervescent antacid tablets (or 3 equal spoonfuls of sugar); 3 clear cups; cold, room-temperature, and warm tap water; thermometer; stopwatch.

1. Measure 200 mL of water into each cup and record each temperature.
2. Drop one tablet into each cup at the same moment and start the timer.
3. Record the time when each tablet is completely dissolved.
4. Repeat if you can, and average the trials.

Record: a data table and a line or bar graph with temperature on the x-axis and time on the y-axis.

Safety: Use warm tap water, not boiling water. Do not swallow the tablets or solutions.

KEY IDEA

Change one independent variable, measure the dependent variable, control everything else, and explain results with CER.

### EXPLAIN IT · WRITE A SCIENTIFIC EXPLANATION

Data: 8°C → 95 s; 21°C → 52 s; 40°C → 24 s

1. Claim: Tablets dissolve faster in warmer water.
2. Evidence: dissolving time dropped from 95 s at 8°C to 52 s at 21°C to 24 s at 40°C.
3. Reasoning: particles in warmer water move faster, so they collide with the tablet more often and break it apart sooner.

REMEMBER

✓ REMEMBER

Independent is changed, dependent is measured, controlled stay the same; explain with CER.

Explanation: A complete explanation connects the evidence to a scientific idea.

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.1, Part A. Then check with your teacher.

1. Plant height is measured after different amounts of fertilizer. Independent variable?
2. Same experiment: dependent variable?
3. Name two controlled variables.

WORDS & MORE

REMEMBER

LESSON 1.2
Week 1 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.2: Earth's Spheres and the Water Cycle

TODAY'S GOAL

I can describe Earth's four spheres and model how energy from the sun and gravity drive the water cycle.

Standards: MS-ESS2-4

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Do the Investigate activity: Water Cycle in a Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–D).

## 1 Get Ready

I WONDER

The water you drank today may once have been in a dinosaur, a cloud, or an ocean. How could that be?

## 2 Learn

Scientists study Earth as a system of four interacting parts, or spheres: the geosphere, hydrosphere, atmosphere, and biosphere. Matter, like water, constantly moves among them. Earth has about the same amount of water today as it had long ago; it just keeps cycling.

WORDS & MORE

WORDS TO KNOW

**geosphere**
The solid rock and soil of Earth.

**hydrosphere**
All of Earth's water.

**atmosphere**
The layer of gases around Earth.

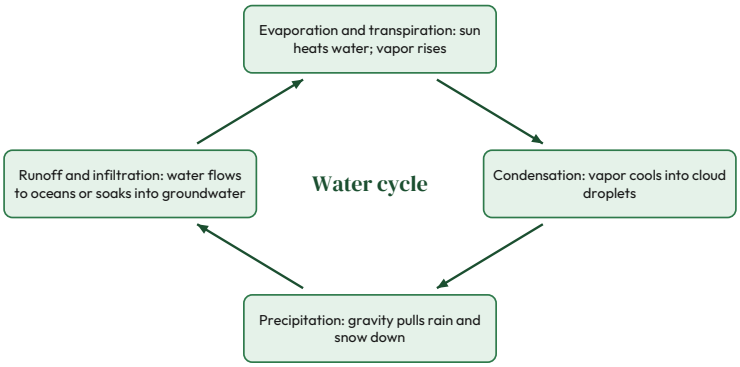
**biosphere**
All living things on Earth.

**transpiration**
The release of water vapor from plant leaves.

**condensation**
The change from water vapor to liquid water.

REMEMBER

WORDS & MORE




Two forces drive the cycle. Energy from the sun evaporates water and powers winds. Gravity pulls precipitation down and makes water flow downhill into rivers, lakes, and oceans, and soak into the ground.

KEY IDEA

Water cycles among Earth's spheres, driven by energy from the sun and by gravity.

INVESTIGATE · WATER CYCLE IN A BAG

Can you model the water cycle?

You need: a zip-top bag; 1/4 cup water with a few drops of blue food coloring; tape; a sunny window; marker.

1. Draw a sun, cloud, and waves on the bag.
2. Pour in the water and seal the bag.
3. Tape it to a sunny window.
4. Observe it after 1 hour, after 1 day, and after 2 days.

Record: observations and a labeled sketch showing evaporation, condensation, and precipitation in the bag.

✓ REMEMBER

Water cycles among Earth's spheres, powered by the sun and gravity.

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.2, Part A. Then check with your teacher.

1. Which sphere includes clouds and air?
2. What process returns water vapor from plants?
3. What force makes rain fall and rivers flow?

REMEMBER

LESSON 1.3
Week 2 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.3: The Carbon and Oxygen Cycles

TODAY'S GOAL

I can explain how carbon and oxygen cycle among living things, the atmosphere, the oceans, and rocks.

Standards: MS-LS2-3

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Do the Investigate activity: Follow a Carbon Atom.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–D).

## 1 Get Ready

I WONDER

Carbon is in your body, in a diamond, in soda bubbles, and in the air. How can one element be in all of these?

## 2 Learn

Carbon is the building block of living things. Carbon atoms are never used up; they cycle. Plants take carbon dioxide from the air during photosynthesis and build sugars. Animals eat plants. Living things release carbon dioxide during cellular respiration, and decomposers release it as

WORDS & MORE

WORDS TO KNOW

**carbon cycle**
The movement of carbon among the air, living things, oceans, and rocks.

**decomposer**
An organism that breaks down dead matter and returns materials to the environment.

**fossil fuel**
Coal, oil, or natural gas formed from ancient living things.

**combustion**
Burning; a reaction with oxygen that releases energy and carbon dioxide.

REMEMBER

they break down dead matter. Oceans absorb and release carbon dioxide too. Over millions of years, some carbon becomes buried as fossil fuels; burning them releases that carbon quickly.

The oxygen cycle is linked to the carbon cycle: photosynthesis releases oxygen, and respiration and combustion use oxygen. Today air is about 21 percent oxygen and about 0.04 percent carbon dioxide.

| Process | Carbon dioxide | Oxygen |
|---|---|---|
| photosynthesis | taken in | released |
| cellular respiration | released | taken in |
| decomposition | released | taken in |
| combustion (burning fuels) | released | taken in |
| ocean absorption | taken in (dissolves in water) | — |

WORDS & MORE

KEY IDEA

Photosynthesis removes carbon dioxide and adds oxygen; respiration, decomposition, and combustion do the opposite.

INVESTIGATE · FOLLOW A CARBON ATOM

What paths can one carbon atom take?

You need: station cards: Atmosphere, Plant, Animal, Decomposer, Ocean, Fossil Fuel; one die.

1. Start at the Atmosphere station.
2. Roll the die; each station card tells where a roll sends you and why (e.g. 1–3: a plant takes you in during photosynthesis).
3. Record each move for 10 rolls.
4. Compare paths with a partner.

Record: your atom's path as a flow chart, with the process that moved it at each arrow.

✓ REMEMBER

Carbon and oxygen cycle through photosynthesis, respiration, decomposition, combustion, and oceans.

REMEMBER

WORDS & MORE

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.3, Part A. Then check with your teacher.

1. Which process removes carbon dioxide from the air?
2. Name two processes that release carbon dioxide.
3. About what percent of air is oxygen?

REMEMBER

LESSON 1.4
Week 2 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.4: The Nitrogen Cycle

TODAY'S GOAL

I can explain how nitrogen moves from the air into living things and back.

Standards: MS-LS2-3

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Do the Investigate activity: Root Nodule Search.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–D).

## 1 Get Ready

I WONDER

Air is about 78 percent nitrogen, and your body needs nitrogen to build proteins. Why can't you just breathe it in and use it?

## 2 Learn

Living things need nitrogen to build proteins and DNA, but most organisms can't use nitrogen gas ($N_2$) directly. It has to be fixed into compounds such as ammonia and nitrates.

WORDS & MORE

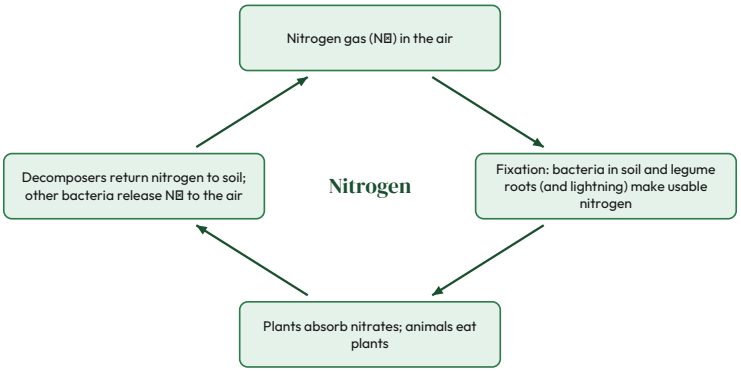
WORDS TO KNOW

**nitrogen fixation**
Changing nitrogen gas into a form plants can use.

**legume**
A plant such as a bean, pea, or peanut that hosts nitrogen-fixing bacteria in its roots.

REMEMBER

WORDS & MORE




Farmers use this cycle. Planting legumes such as soybeans or peanuts adds usable nitrogen to the soil. Peanut farmer and scientist George Washington Carver taught Southern farmers to rotate cotton with peanuts and other legumes to restore worn-out soil.

KEY IDEA

Bacteria fix nitrogen gas into usable forms; plants absorb it, animals eat plants, and decomposers and bacteria return it.

INVESTIGATE · ROOT NODULE SEARCH

Where do nitrogen-fixing bacteria live?

You need: a bean or pea plant (or photos from your teacher); magnifying glass; water.

1 Gently dig up the plant and rinse the soil from its roots.

2 Look for small round bumps (nodules) on the roots.

3 Sketch the roots and nodules.

Record: a labeled sketch and an explanation of what the bacteria in the nodules do.

✓ REMEMBER

Bacteria fix nitrogen so plants and animals can use it; decomposers return it.

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.4, Part A. Then check with your teacher.

1. What percent of air is nitrogen?
2. What does nitrogen fixation do?
3. Why do farmers plant legumes?

REMEMBER

LESSON 1.5
Week 3 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.5: Photosynthesis

TODAY'S GOAL

I can explain how plants use light energy to make sugar and oxygen from carbon dioxide and water.

Standards: MS-LS1-6

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Do the Investigate activity: Floating Leaf Disks.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–D).

## 1 Get Ready

I WONDER

A huge oak tree grows from a tiny acorn. Where does all that mass come from?

## 2 Learn

Surprisingly, most of a tree's mass comes from the air! In photosynthesis, chloroplasts in leaf cells use light energy captured by chlorophyll to combine carbon dioxide from the air with water from the soil. They produce glucose, which the plant uses for energy and to build its body, and release oxygen.

WORDS & MORE

WORDS TO KNOW

**photosynthesis**
The process plants use to turn light energy, carbon dioxide, and water into sugar and oxygen.

**chloroplast**
The part of a plant cell where photosynthesis happens.

**chlorophyll**
The green pigment that captures light energy.

**glucose**
A sugar made in photosynthesis that stores energy.

REMEMBER

WORDS & MORE

| | |
|---|---|
| Word equation | carbon dioxide + water —(light energy)→ glucose + oxygen |
| Chemical equation | $6CO_2 + 6H_2O \xrightarrow{light} C_6H_{12}O_6 + 6O_2$ |

Photosynthesis changes light energy into chemical energy stored in sugar. Almost every food chain on Earth begins with this process.

KEY IDEA

Photosynthesis: carbon dioxide + water + light energy → glucose + oxygen, in chloroplasts.

INVESTIGATE · FLOATING LEAF DISKS

Does light intensity affect the rate of photosynthesis?

You need: fresh spinach leaves; a straw or hole punch; a plastic syringe (no needle); baking soda solution (1/8 tsp in 300 mL water) with 1 drop dish soap; 2 clear cups; a lamp.

1. Punch 20 disks from the leaves.
2. Put 10 disks in the syringe with solution, push out the air, cover the tip, and pull the plunger to remove gas from the leaves until the disks sink. Repeat for the other 10.
3. Put 10 sunk disks in a cup under the lamp and 10 in a cup in a dark place.
4. Count floating disks every minute for 15 minutes.

Record: a line graph of floating disks vs time for both cups, and a CER explaining the difference (oxygen produced in photosynthesis makes disks float).

✓ REMEMBER

Plants turn carbon dioxide and water into glucose and oxygen using light.

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.5, Part A. Then check with your teacher.

1. Where does most of a tree's mass come from?
2. What are the products of photosynthesis?
3. What pigment captures light?

REMEMBER

LESSON 1.6
Week 3 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.6: Cellular Respiration

TODAY'S GOAL

I can explain how cells release energy from food through cellular respiration.

Standards: MS-LS1-7

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Do the Investigate activity: Yeast Balloons.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–D).

## 1 Get Ready

I WONDER

You ate breakfast hours ago. How is that food powering your brain right now?

## 2 Learn

All organisms, including plants, need energy. In cellular respiration, mitochondria in cells break down glucose with oxygen, releasing energy the cell can use. Carbon dioxide and water are given off as waste.

WORDS & MORE

WORDS TO KNOW

**cellular respiration**
The process cells use to release energy from glucose using oxygen.

**mitochondria**
Cell parts where cellular respiration releases energy.

**fermentation**
A process that releases energy from sugar without oxygen.

REMEMBER

| | |
|---|---|
| Word equation | glucose + oxygen → carbon dioxide + water + energy |
| Chemical equation | $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy |

Notice that respiration is roughly the reverse of photosynthesis. The products of one are the reactants of the other, so matter cycles between plants and animals while energy flows from the sun, through living things, and is eventually lost as heat.

Some organisms, like yeast, can also release energy without oxygen through fermentation, giving off carbon dioxide. That's why bread dough rises.

WORDS & MORE

KEY IDEA

Respiration: glucose + oxygen → carbon dioxide + water + energy, in mitochondria. It is the reverse of photosynthesis.

INVESTIGATE · YEAST BALLOONS

Does yeast need sugar to release carbon dioxide?

You need: 2 empty water bottles; 2 balloons; 2 packets of dry yeast; warm water; 2 tablespoons sugar.

1. Add 1 cup warm water and 1 packet yeast to each bottle.
2. Add sugar to bottle A only. Swirl both.
3. Stretch a balloon over each bottle's mouth.
4. Measure each balloon's circumference every 10 minutes for 30 minutes.

Record: a data table and a graph; explain what filled the balloon and why one grew more.

✓ REMEMBER

Cells release energy from glucose with oxygen; respiration and photosynthesis cycle matter.

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.6, Part A. Then check with your teacher.

1. Where does cellular respiration happen?
2. What gas filled the balloon?
3. Do plants carry out respiration?

REMEMBER

LESSON 1.7
Week 4 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.7: Food Webs and Energy Pyramids

TODAY'S GOAL

I can model how energy flows through a food web and explain why less energy reaches each higher level.

Standards: MS-LS2-3

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Do the Investigate activity: Build a Food Web.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–D).

## 1 Get Ready

I WONDER

Why are there far more grasshoppers than hawks in a field?

## 2 Learn

Energy enters most ecosystems as sunlight captured by producers. Primary consumers (herbivores) eat producers, secondary consumers eat herbivores, and so on. Decomposers break down dead organisms at every level. Arrows in a food chain point in the direction energy flows:

WORDS & MORE

WORDS TO KNOW

**producer**
An organism that makes its own food, usually by photosynthesis.

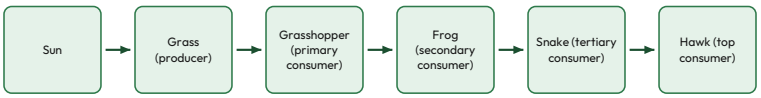
**consumer**
An organism that eats other organisms.

**food web**
A diagram of all the feeding relationships in an ecosystem.

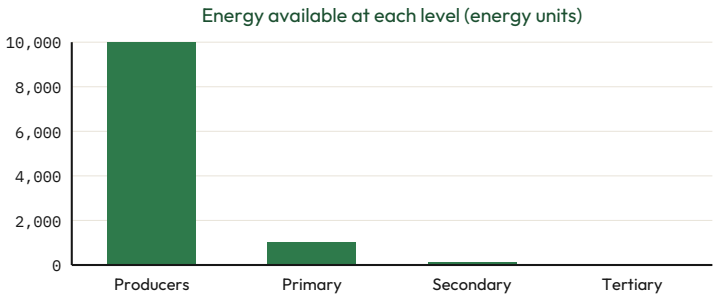
**trophic level**
A feeding level in a food chain or energy pyramid.

REMEMBER

from the eaten to the eater.




At each level, organisms use most of their energy to live, move, and stay warm, and much is lost as heat. On average only about 10 percent of the energy at one level is stored in the next. That's why an energy pyramid narrows at the top and why there are fewer top predators.




KEY IDEA

Energy flows from producers to consumers; only about 10 percent passes to each higher level, while matter is recycled by decomposers.

INVESTIGATE · BUILD A FOOD WEB

How are organisms in a forest ecosystem connected?

You need: cards: oak tree, grass, acorn weevil, grasshopper, mouse, rabbit, deer, frog, snake, hawk, fox, mushroom (decomposer); yarn or a pencil.

1 Arrange the cards on a large sheet of paper.

2 Draw arrows from each food to the organism that eats it.

3 Label producers, consumers, and the decomposer.

4 Remove the mouse. Predict two effects on the web.

Record: your food web and your predictions.

✓ REMEMBER

Energy flows up through feeding levels with about 10 percent passed on; matter recycles.

WORDS & MORE

REMEMBER

WORDS & MORE

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.7, Part A. Then check with your teacher.

1. Which way do food-chain arrows point?
2. If grass has 20,000 units, about how much reaches grasshoppers?
3. Why are top predators few?

REMEMBER

LESSON 1.8
Week 4 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

# Lesson 1.8: Biodiversity and Ecosystem Services

TODAY'S GOAL

I can explain why biodiversity matters and evaluate ways to protect ecosystem services.

Standards: MS-LS2-5

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Do the Investigate activity: Biodiversity Survey.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–D).

## 1 Get Ready

I WONDER

What would happen to your breakfast if all the bees disappeared?

## 2 Learn

Biodiversity is the variety of species, genes, and ecosystems in a place. Ecosystems with high biodiversity are usually more stable: if one species declines, others can fill its role.

WORDS & MORE

WORDS TO KNOW

**biodiversity**
The variety of living things in an area.

**ecosystem services**
Benefits people get from healthy ecosystems, such as clean water and pollination.

**pollinator**
An animal, such as a bee, that carries pollen between flowers.

REMEMBER

| Ecosystem service | Example |
|---|---|
| pollination | Bees, butterflies, and bats pollinate many fruits, nuts, and vegetables. |
| clean water | Wetlands filter pollutants as water flows through them. |
| flood control | Wetlands and forests soak up heavy rain like sponges. |
| healthy soil | Decomposers and earthworms recycle nutrients. |
| food and materials | Fish, timber, and medicines come from ecosystems. |

WORDS & MORE

EXPLAIN IT · EVALUATE A DESIGN SOLUTION

A town wants to reduce flooding. Option A: build a concrete channel. Option B: restore a wetland upstream.

1. Criteria: reduce flooding, cost, protect biodiversity, clean water.
2. A: moves water fast but removes habitat and doesn't filter water.
3. B: soaks up and slows water, filters pollutants, and provides habitat, but needs land.

Explanation: Option B meets more criteria, especially biodiversity and clean water, though land and cost are constraints.

KEY IDEA

Biodiversity keeps ecosystems stable and provides services people depend on; good design solutions protect it.

INVESTIGATE · BIODIVERSITY SURVEY

How many kinds of living things live in a small area?

You need: 4 sticks and string to mark a 1 m × 1 m square; magnifying glass; paper and a pencil.

1. Mark a square in a lawn and another in a less-mowed area (garden edge, field, or woods).
2. Count the different kinds of plants and animals in each square for 10 minutes.
3. Compare the two squares.

Record: a table of species counts and an explanation of which area has more biodiversity and why.

✓ REMEMBER

Biodiversity stabilizes ecosystems and provides services worth protecting.

REMEMBER

WORDS & MORE

## 3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.8, Part A. Then check with your teacher.

1. What is biodiversity?
2. Name two ecosystem services.
3. Why is a diverse ecosystem more stable?

UNIT 1 REVIEW

# Unit 1 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

| Lesson | Remember... |
|---|---|
| 1.1 | Independent is changed, dependent is measured, controlled stay the same; explain with CER. |
| 1.2 | Water cycles among Earth's spheres, powered by the sun and gravity. |
| 1.3 | Carbon and oxygen cycle through photosynthesis, respiration, decomposition, combustion, and oceans. |
| 1.4 | Bacteria fix nitrogen so plants and animals can use it; decomposers return it. |
| 1.5 | Plants turn carbon dioxide and water into glucose and oxygen using light. |
| 1.6 | Cells release energy from glucose with oxygen; respiration and photosynthesis cycle matter. |
| 1.7 | Energy flows up through feeding levels with about 10 percent passed on; matter recycles. |
| 1.8 | Biodiversity stabilizes ecosystems and provides services worth protecting. |

# Glossary

| | |
|---|---|
| atmosphere | The layer of gases around Earth. |
| biodiversity | The variety of life in an area. |
| biosphere | All living things on Earth. |
| carbon cycle | The movement of carbon among the air, living things, oceans, and rocks. |
| cellular respiration | Glucose + oxygen → $CO_2$ + water + energy. |
| chlorophyll | The green pigment that captures light energy. |
| chloroplast | The part of a plant cell where photosynthesis happens. |
| combustion | Burning; a reaction with oxygen that releases energy and carbon dioxide. |
| condensation | The change from water vapor to liquid water. |
| consumer | An organism that eats other organisms. |
| controlled variables | Conditions kept the same so the test is fair. |
| decomposer | An organism that breaks down dead matter and returns materials to the environment. |
| dependent variable | The variable that is measured. |
| ecosystem services | Benefits people get from healthy ecosystems, such as clean water and pollination. |
| fermentation | A process that releases energy from sugar without oxygen. |
| food web | A diagram of all the feeding relationships in an ecosystem. |
| fossil fuel | Coal, oil, or natural gas formed from ancient living things. |
| geosphere | The solid rock and soil of Earth. |
| glucose | A sugar made in photosynthesis that stores energy. |
| hydrosphere | All of Earth's water. |
| hypothesis | A testable prediction that explains how one variable affects another. |
| independent variable | The variable that is changed. |
| legume | A plant such as a bean, pea, or peanut that hosts nitrogen-fixing bacteria in its roots. |
| mitochondria | Cell parts where cellular respiration releases energy. |
| nitrogen fixation | Changing $N_2$ into usable forms. |
| photosynthesis | Light + $CO_2$ + water → glucose + oxygen. |

| | |
|---|---|
| pollinator | An animal, such as a bee, that carries pollen between flowers. |
| producer | An organism that makes its own food, usually by photosynthesis. |
| transpiration | The release of water vapor from plant leaves. |
| trophic level | A feeding level in a food chain or energy pyramid. |

# Grade 7 Science

Student Textbook

A full year of Grade 7 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Textbook teaches every lesson: the big idea, key vocabulary, models and examples, and a short set of practice problems to check understanding. It pairs with the Student Workbook and the Teacher Edition.

## INSIDE THIS YEAR


**01** Cycles of Matter and Energy


## IN THIS BOOK

- Clear lesson goals: “I can…” statements
- Key vocabulary with student-friendly definitions
- Models, examples, and Key Idea summaries
- A study guide and glossary for every unit

## THE COMPLETE SET




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